

ANSH SINGH

Backend Engineer | Python, Django, FastAPI | GenAI & LLM Systems

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SUMMARY

Backend Engineer with 1.5 years of experience building and scaling production systems for fintech and health-tech platforms using **Python, Django, FastAPI, and Ruby on Rails**. Shipped 25+ production APIs serving loan-origination workflows for **Jio Finance**, cutting manual operations by 60% and approval turnaround from 2 days to 3 hours. Hands-on with **Kafka, Redis, Celery, PostgreSQL**, and event-driven architecture, plus applied GenAI: **RAG pipelines, embeddings, vector databases, and LLM integration** (OpenAI, Gemini, local Llama 3 via Ollama).

TECHNICAL SKILLS

Languages: Python, SQL, C++, Ruby, Bash

Backend: Django, Django REST Framework, FastAPI, Ruby on Rails, REST API Design, JWT Authentication, RBAC, API Versioning, WebSockets, Rate Limiting, Microservices

Data & Infrastructure: PostgreSQL, Redis, Kafka, Celery, RabbitMQ, Database Indexing, Query Optimization, Docker, Git, GitHub Actions (CI/CD), Linux, Nginx, Gunicorn, AWS (EC2, S3, RDS), Prometheus & Grafana (Monitoring), pytest, OpenAPI/Swagger

Generative AI: LLM Integration (OpenAI, Gemini, Llama 3/Ollama), Retrieval-Augmented Generation (RAG), Embeddings, Vector Databases (pgvector, FAISS), LangChain, LangGraph, Prompt Engineering, AI Agents

Core CS: Data Structures & Algorithms, OOP, Operating Systems, DBMS, System Design

EXPERIENCE

Backend Developer

Jan 2025 – Jul 2026

Bits Flow Technology Pvt. Ltd.

India

- **Finity (Jio Finance, PFL) — Loan Counsellor E2E Module:** Engineered and owned **25+ production Django REST APIs** with 99.9% uptime, implementing JWT authentication, role-based access control, and rate limiting for a fintech loan-origination platform.
- Led backend development of the Counsellor Module end-to-end (data models, APIs, admin workflows), improving loan tracking and counsellor operational efficiency by **40%**.
- Integrated third-party KYC/verification services via **Kafka-based asynchronous pipelines** with retry and dead-letter handling, reducing customer onboarding time by **35%**.
- **Business Rule Engine (BRE):** Designed a configurable rule-based decision engine automating **15+ loan eligibility checks**, reducing manual credit-review effort by **60%** and approval turnaround from **2 days to 3 hours** using distributed task queues and real-time data flows.
- Built admin-driven rule configuration with **Redis caching** for low-latency (sub-100ms) rule evaluation, enabling business teams to update eligibility logic without code deployments.
- **PranaAI (Health-Tech):** Architected and delivered a production-ready **Ruby on Rails** MVP in 2 months — wellness tracking, activity logging, authentication, and AI-driven recommendation pipelines — collaborating directly with stakeholders on API contracts and data models.

PROJECTS

LedgerFlow — Real-time Loan Processing Platform | *FastAPI, PostgreSQL, Kafka, Redis, Celery, Docker* | GitHub

- Building an open-source fintech backend: configurable rule engine with Redis-cached admin-editable eligibility rules, Kafka verification pipelines with dead-letter handling, idempotency keys on all write endpoints, and Prometheus/Grafana observability with published Locust load-test results.

MyPdfAgent | *Python, FastAPI, OCR, Ollama (Llama 3), Embeddings* | GitHub

- Built an AI-powered PDF reconstruction system that reorders shuffled multi-page documents using OCR, embedding similarity, and LLM reasoning; hybrid AI + rule-based approach cut reconstruction time by **70%**.
- Ran fully **local LLM inference with Ollama (Llama 3)** to guarantee document privacy — no data leaves the machine.

PythonGeneratorAgent | *FastAPI, Google Gemini 2.0, Function Calling* | GitHub

- Developed an AI coding assistant with FastAPI backend: AI-driven Python code generation, sandboxed execution, real-time file operations, and function-call visualization via **Gemini 2.0 function calling**.

AI Resume Screening Platform | *FastAPI, Embeddings, Vector Search, Event-Driven Pipelines*

- Built a SaaS platform for automated resume screening and candidate-job matching using **vector similarity search** over embeddings, with event-driven ingestion pipelines for large-scale resume processing.

EDUCATION

Thapar Institute of Engineering and Technology

2021 – 2025

B.E. Mechanical Engineering — CGPA: 7.4

Patiala, India

- Coursework: Data Structures, Algorithms, Machine Learning, Deep Learning, OOP, Operating Systems

ACHIEVEMENTS & CERTIFICATIONS

- Solved **200+ Data Structures & Algorithms** problems across LeetCode, GeeksforGeeks, and InterviewBit.
- Python for Data Science and Machine Learning Bootcamp (Udemy).